College of Engineering Muttathara
Department of Electronics and Communication Engineering

GROUP 1

IOT ENABLED NOISE MONITORING SYSTEM

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INTRODUCTION

Noise pollution has become a growing environmental concern in modern urban life, with sources such as traffic, industrial activity, and crowded public spaces contributing to increased noise levels. Prolonged exposure to high noise levels can lead to serious health issues. In sensitive environments maintaining a quiet atmosphere is crucial for comfort, concentration, and well-being.

This project aims to address the issue of noise pollution .The system continuously monitors sound levels and alert them.

By implementing this system, the project helps promote noise awareness and encourages proactive control of sound disturbances in noise-sensitive areas.

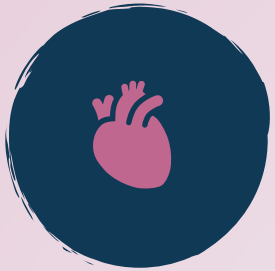
ABSTRACT

Noise pollution is a significant environmental issue affecting health and productivity in classrooms, industries, and urban areas. Prolonged exposure to high noise levels can cause stress, hearing impairments, and reduced concentration. To address this, an IoT- based noise monitoring system is developed using ESP32 to detect, analyze, and alert users about excessive noise levels in real time. The system integrates a sound sensor to continuously measure noise levels in decibels (dB), with real-time data displayed on an LCD screen. If noise exceeds a predefined threshold, a buzzer rings, and an alert is sent via the Blynk app for remote monitoring. In classroom environments, when the noise crosses the set limit, the ESP32-CAM captures an image, helping to identify the source of disturbance. Additionally, the system logs noise data and represents it through visual charts, allowing trend analysis for better noise management. With Wi-Fi connectivity, users can remotely monitor noise levels, making it easier to implement corrective measures. This IoT-based noise monitoring system offers an efficient and automated solution for tracking, analyzing, and controlling noise pollution, improving classroom discipline, workplace safety, and overall noise management.

NOISE POLLUTION AND ITS EFFECTS

Noise pollution refers to any unwanted or excessive sound that can negatively impact human and animal health, as well as environmental quality.Noise pollution is a leading preventable cause of hearing loss, affecting an estimated 12% of the global population. Worldwide, 16% of disabling hearing loss in adults is attributed to occupational noise.

HEARING LOSS



HEART PROBLEMS

AFFECTS PREGNANT WOMEN

SLEEP DISTURBANCES

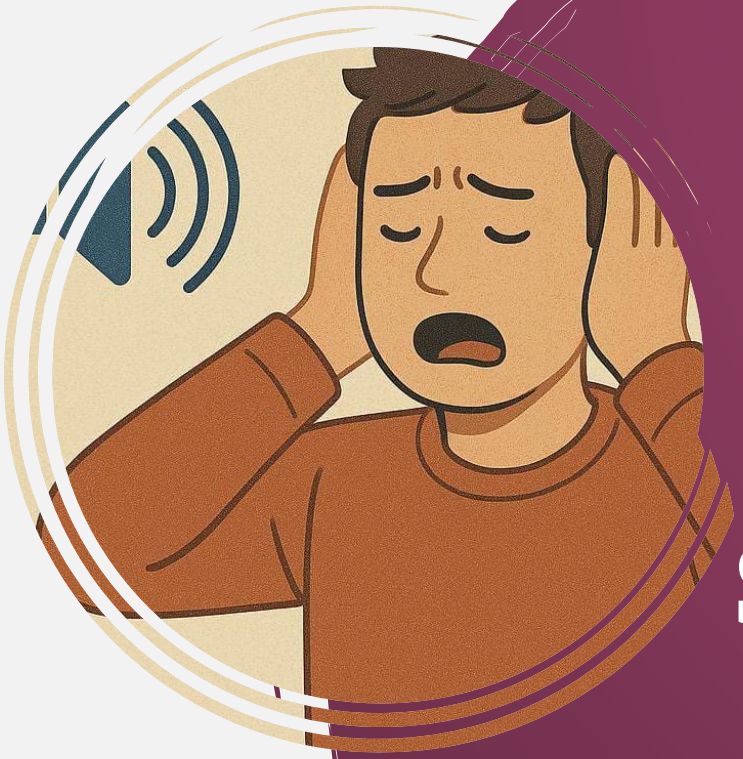
CAUSE STRESS AND HEADACHE

- Noise pollution can impact people of all ages, but it's especially harmful to the very young and the very old. The World Health Organization (WHO) says that sound levels above 55 decibels are harmful to human health.
- A noise beyond 75 dB can be painful and will affect the person severely. It is impossible to see the danger posed by noise pollution. On land and under the sea, you can't see it, but it still exists. Humans and other organisms can be affected adversely by noise pollution if it is an unwanted or disturbing sound.

OBJECTIVE

- Detect and monitor noise levels using a sound sensor.
- Alert users with a buzzer when noise exceeds a set threshold.
- Display status messages on an LCD screen.
- Capture images during noise events using ESP32-CAM.
- Provide a simple, low-cost IoT-based solution for noise monitoring.
- Promote awareness of noise pollution in sensitive areas.

PROBLEM STATEMENT



- Over 20% of the global population is at risk of hearing loss due to long-term exposure to high noise levels (WHO).
- Exposure to noise levels above 85 dB for extended periods can lead to permanent hearing damage.
- Noise pollution contributes to 48% of cardiovascular issues in urban areas due to stress and increased blood pressure (European Environment Agency).

- Around 40% of students report difficulty concentrating in noisy classrooms, affecting learning outcomes.
- Sleep disturbance caused by night-time noise affects up to 30% of adults, leading to fatigue and reduced performance.

LITERATURE REVIEW

TITLE & NAME OF AUTHOR	JOURNAL NAME & YEAR	OVERVIEW	POSITIVE ASPECTS	LIMITATIONS
IoT-Based Air and Noise Pollution Monitoring System Author:Tanuja Borate	Open Access Journal of Science and Engineering Year:2019	This system monitors air and noise pollution using IoT-enabled sensors. It features real-time data collection and remote monitoring via a cloud-based platform.	• Simultaneous monitoring of air and noise pollution. • Cloud-based system for remote accessibility.	Does not address challenges in multi-pollutant analysis and urban scalability.
IoT Based Real-Time Noise Mapping System for Urban Sound Pollution Study Author:Sakib Ahmed, Touseef Saleh Bin Ahmed, Sumaiya Jafreen, Jannatul Tajrin, Jia Uddin	ARXIV Year:2020	This paper describes the development of a system that enables real-time data visualization regarding sound intensity using multiple node devices connected through the internet.	• Facilitates real-time noise mapping across multiple locations. • Employs IoT and web technologies for data visualization.	The system's scalability and data accuracy in high-density urban areas require further exploration.

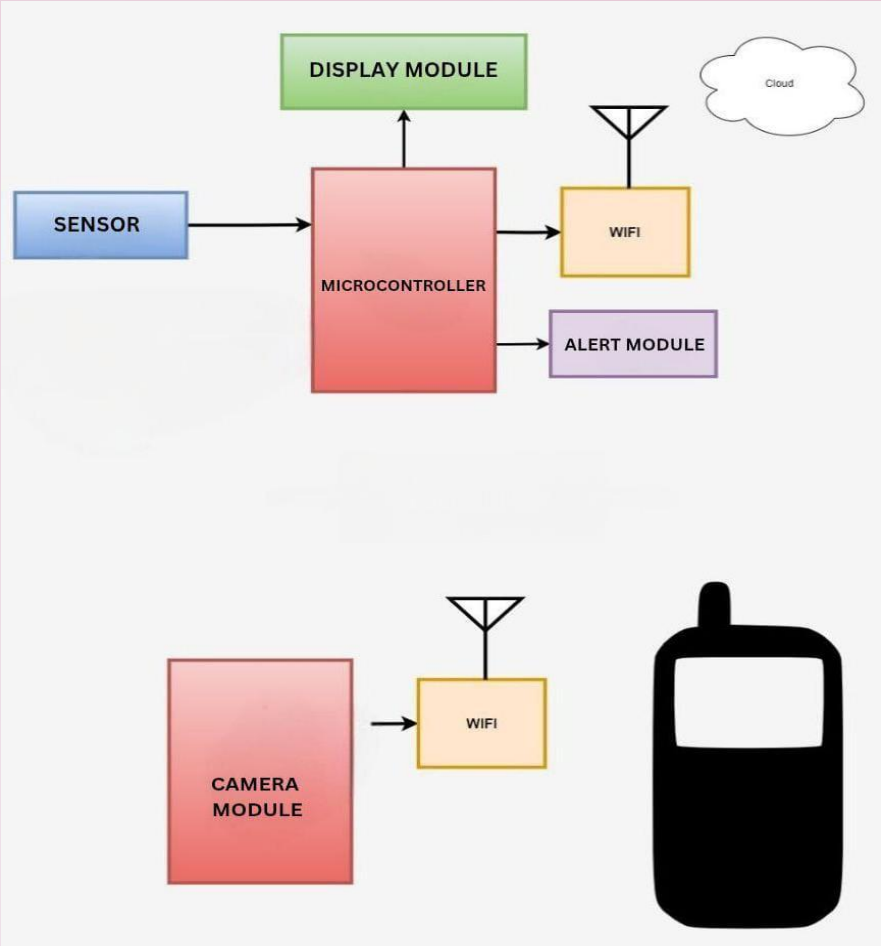
TITLE & NAME OF AUTHOR	JOURNAL NAME & YEAR	OVERVIEW	POSITIVE ASPECTS	LIMITATIONS
IoT-Enabled Acoustic Sensing Platform for Noise Pollution Monitoring Author:Gunatilaka Dolvara	Proceedings of the IEEE 12th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON). Year:2021	A low-cost acoustic sensing system for real-time noise monitoring using IoT. It integrates NB-IoT devices for sound measurement, with backend systems for data management and visualization.	• Affordable and scalable system design. • Reliable performance with low latency and energy consumption.	Limited deployment in dense urban environments with variable noise sources.
Design of Low-Cost Wireless Noise Monitoring Sensor Unit Based on IoT Concept Author:Anachkova Maja	Journal of Vibroengineering Year:2021	Proposes a wireless noise sensor design for continuous environmental monitoring as part of the "Smart City" concept. It incorporates IoT and data visualization techniques for efficient noise data collection.	• Cost-effective and suitable for smart city implementations. • Emphasizes granularity in data collection.	Challenges in noise event classification and high-density deployment.

TITLE & NAME OF AUTHOR	JOURNAL NAME & YEAR	OVERVIEW	POSITIVE ASPECTS	LIMITATIONS
Air and noise pollution monitoring system Author:Amulya Gulati Amit Prakash Jayashree. A K. P. Suresh	International Journal of Science and Research (IJSR) Year:2022	The project utilizes IoT and smart sensors to monitor air and noise pollution in real time, providing a cost-effective solution for sustainable environmental management.	• a wireless sensor network that automatically transmits, processes, analyses, and visualises data • The IOT concept is useful in achieving real-time monitoring of sensor data.	Poor sensory data quality requires preprocessing. Lack of robust classification methods for noisy and big data.
IoT Cloud-Based Noise Intensity Monitoring System Author:Fatema Tashfat	Indonesian Journal of Electrical Engineering and Computer Science Year:2023	The system leverages Arduino IoT Cloud for real-time noise data collection and management. It is designed for portability and accessibility through web and mobile applications, aiming to enhance urban living by monitoring noise pollution in cities like Dhaka.	• Portable and user-friendly. • Effective for urban areas with limited monitoring infrastructure.	Focused primarily on specific urban scenarios without addressing scalability.

TITLE & NAME OF AUTHOR	JOURNAL NAME & YEAR	OVERVIEW	POSITIVE ASPECTS	LIMITATIONS
Smart Noise Pollution Monitoring System Author:R. Subraja, T. Meyyapan, A. Padmapriya, S. Santhosh Kumar	World Journal of Advanced Engineering Technology and Sciences Year:2024	This IoT-based system monitors real-time noise levels, processes data via a microcontroller, and uploads it to a server for analysis and alerts. Designed for scalability, it is ideal for sensitive areas like schools, hospitals, and residential zones.	• Real-time noise monitoring with precise data collection. • Scalable for larger areas and adaptable to various settings.	Limited discussion on sensor calibration for diverse environments.Lack of detailed evaluation in highly noisy urban settings.
Design of a Noise Detection System Using Arduino Author:C. N. Vanitha, K. L. Sridhar, R. Dhivakar	GIETU JOURNAL Year:2024	This research presents an affordable and adaptable automation solution for detecting noise pollution in urban areas, employing an IoT paradigm to detect human speech and send personalized alarm messages via Arduino.	• Focuses on urban noise pollution, particularly human speech. • Incorporates IoT for real-time monitoring and alerts.	The system's accuracy in diverse environmental conditions is not extensively discussed.

SYSTEM OVERVIEW

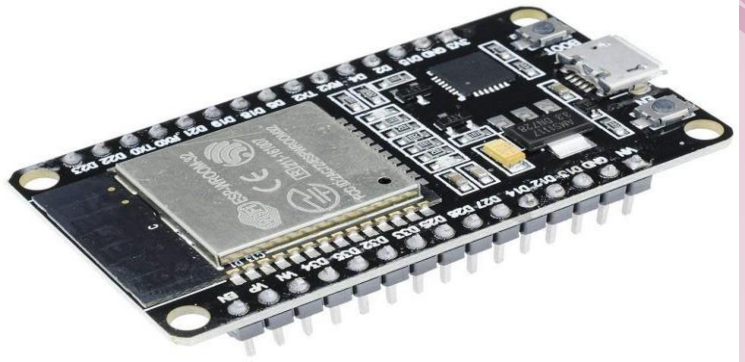
- Microcontroller (e.g., Arduino/ESP32): Acts as the brain of the system, handling sensor inputs, processing data, and controlling outputs.
- Sound Sensor (e.g., microphone module like MAX4466 or KY-037): Captures ambient sound and converts it into analog signals.
- Display Unit (e.g., OLED/LCD): Shows the measured sound levels in decibels (dB).
- Wireless Module (optional - WiFi/Bluetooth): For remote monitoring or sending alerts.
- Camera Module (e.g., ESP32-CAM): Captures real-time images or video. It includes an onboard microcontroller (ESP32) with Wi-Fi capability, allowing for both image processing and wireless data transmission.
- Alert Module (e.g., Buzzer): Provides audible alerts or notifications when certain conditions are met (e.g., when noise is detected). It is controlled by the microcontroller and activated through digital signals.



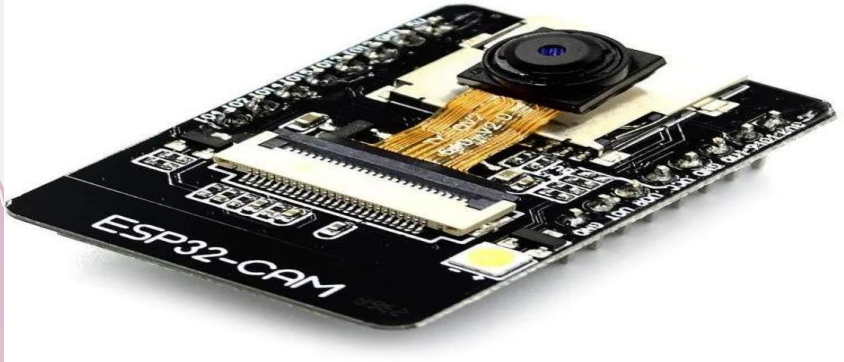
COMPONENTS

The main hardware components of our project are

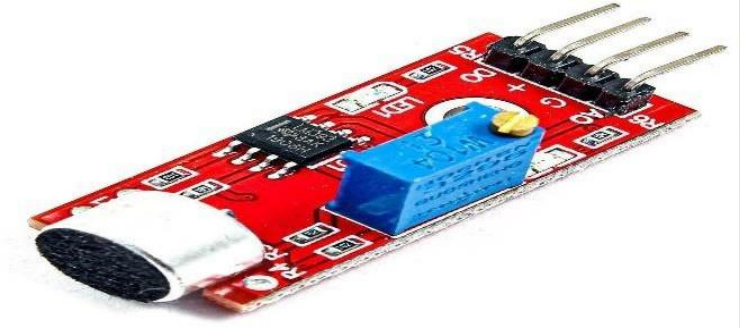
- ESP32
- ESP32 CAM
- Sound Sensor
- LCD Display
- Buzzer
- Power Supply Board
- Connecting wires



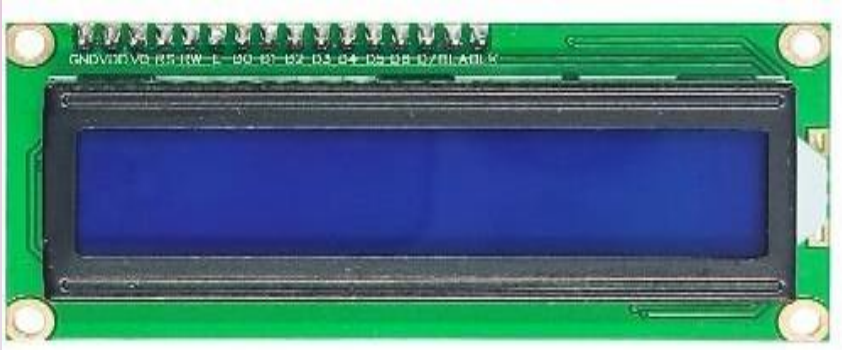
ESP32



ESP32 CAM



Sound Sensor



LCD Display

Buzzer

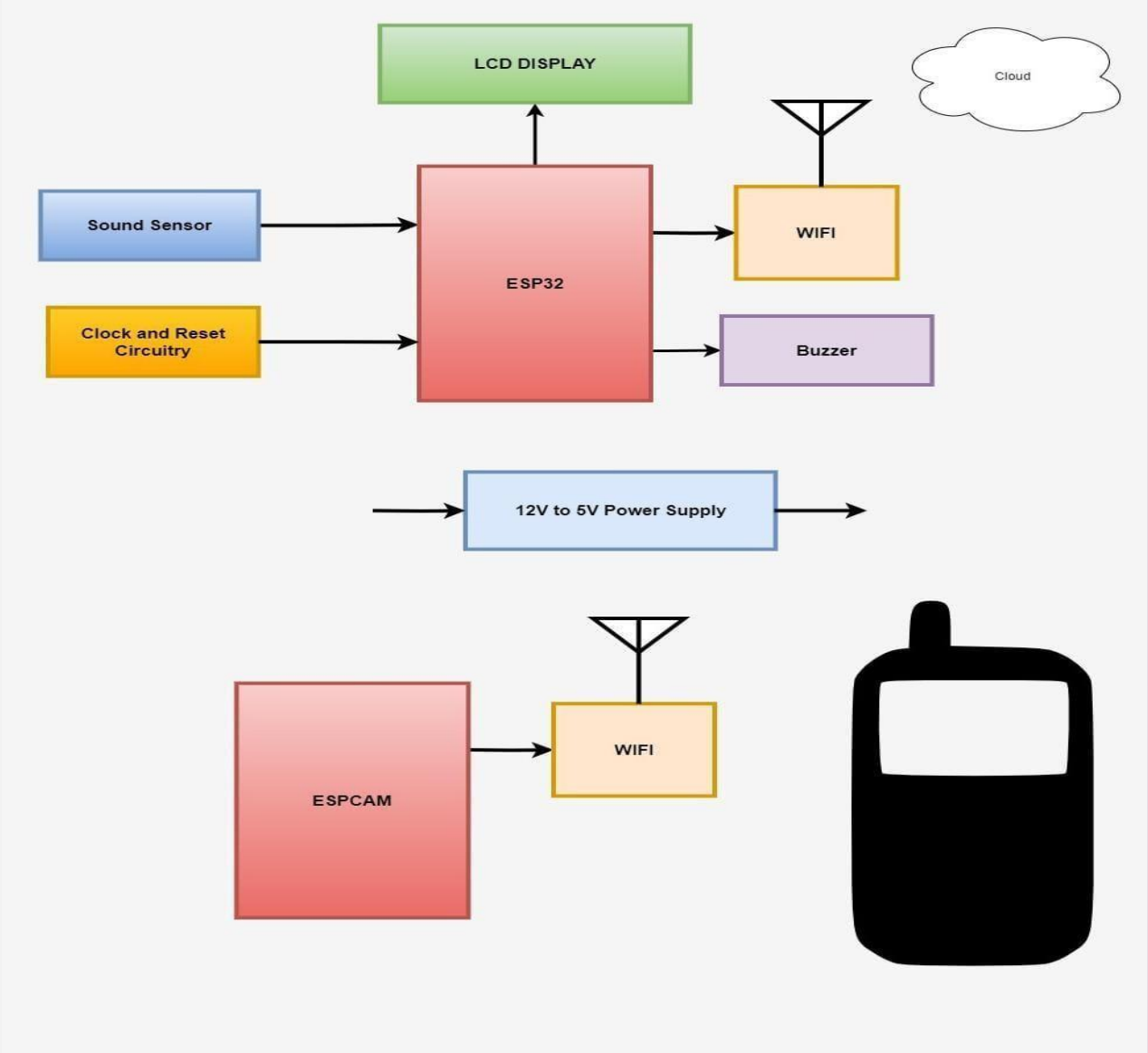
Connecting wires

DESIGN

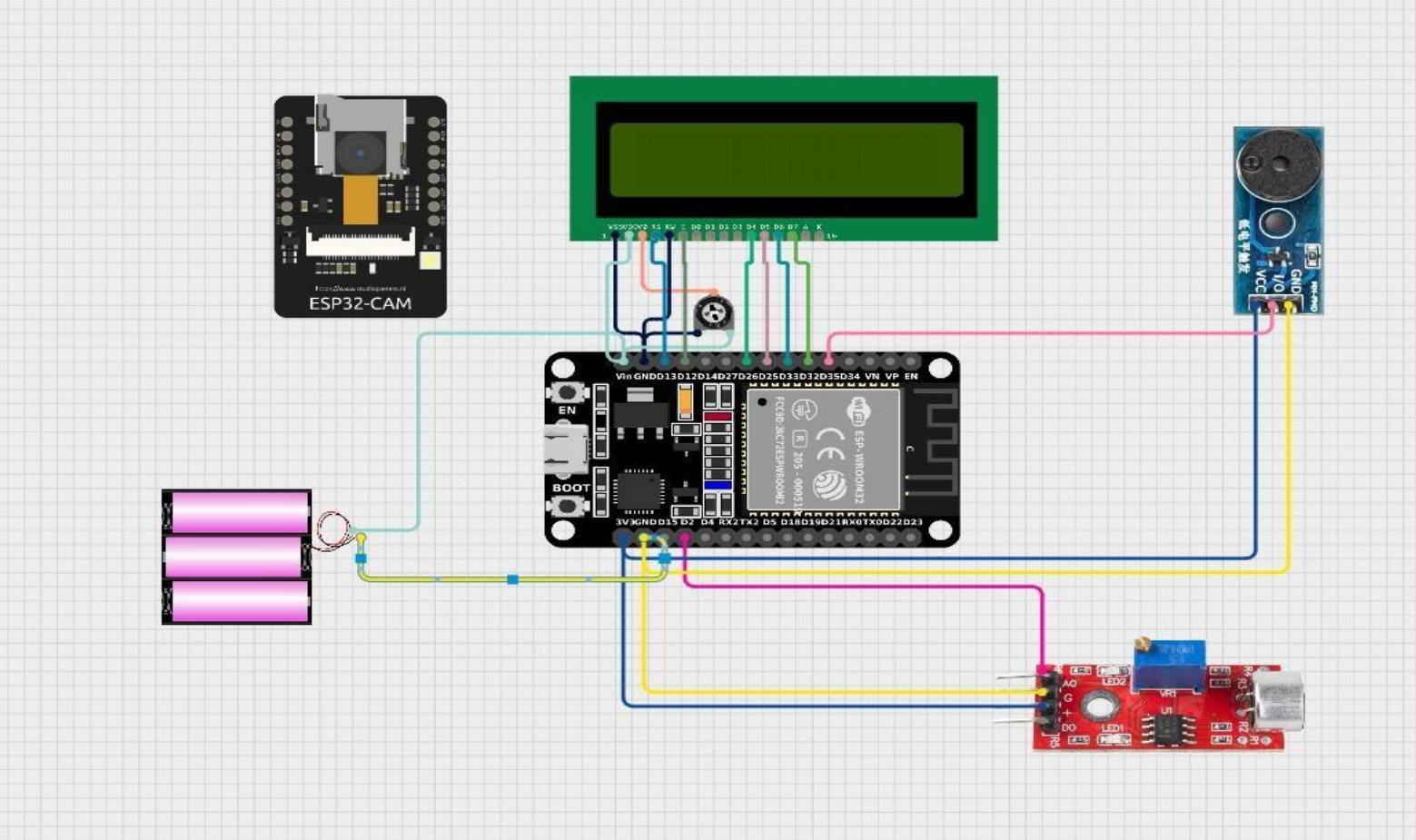
The design of the IoT-enabled noise monitoring system consists of both hardware and software components working together to detect, analyze, and manage noise levels in real time. The ESP32 microcontroller serves as the core unit, processing data from a sound sensor that continuously measures noise levels in decibels (dB). A LCD display shows real-time noise readings, while a buzzer is triggered when noise exceeds a predefined threshold in industries. For classroom monitoring, an ESP32-CAM captures images when noise levels surpass the limit, helping identify sources of disturbance. The system uses Wi-Fi communication to send alerts via the Blynk IoT platform, enabling remote monitoring.

Additionally, noise data is logged and visualized through graphs for trend analysis. The entire system is programmed using Arduino IDE and designed to enhance classroom discipline and industrial safety through automated noise control and monitoring.

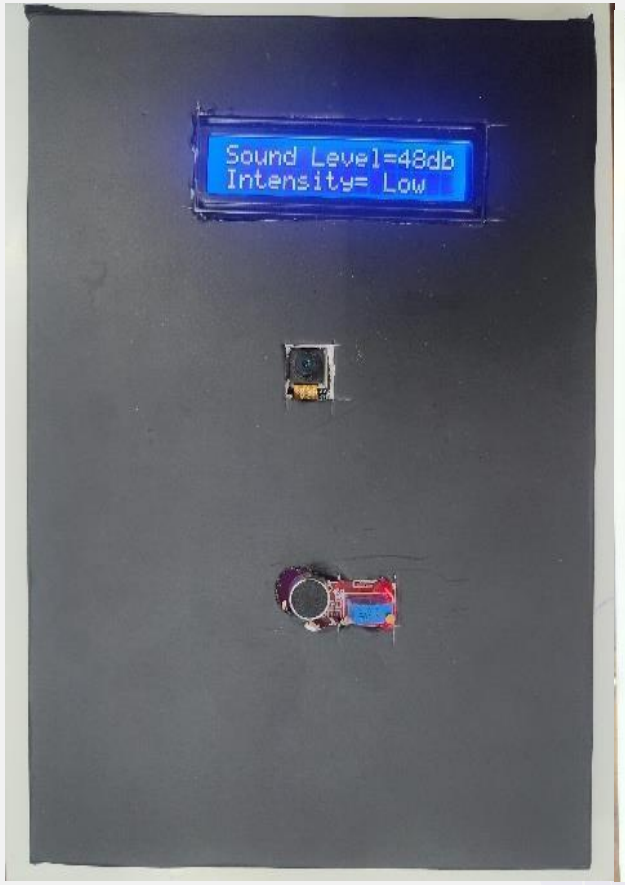
BLOCK DIAGRAM



CIRCUIT DIAGRAM



RESULT


Sound Level=48db
Intensity= Low

Sound Level=51db
Intensity=Medium

Sound Level=99db
Intensity= High

The IoT Enabled Noise Monitoring System was successfully designed and implemented using ESP32 and ESP32-CAM. The system effectively detects excessive noise levels using a sound sensor, displays alerts on an LCD, and triggers a buzzer when the noise crosses a set threshold.

Additionally, the ESP32-CAM captures images or video during noisy events and transmits them via WiFi for remote monitoring. Despite minor challenges related to sensor calibration, power stability, and WiFi connectivity, the system performed reliably in real-time scenarios. It proves to be a useful tool for monitoring and controlling noise pollution in environments such as industries,classrooms,libraries,offices etc.

PLAN OF ACTION

PLAN	STATUS	PERIOD
Identification of Problem	Completed	January
Formation of Idea	Completed	January
Coming to a Solution	Completed	January-February
Making Idea into Project	Completed	February
Purchased Required Components	Completed	March
Prototyping	Completed	March-April
Testing and Implementation	Completed	March-April

CHALLENGES FACED

- Camera Performance: ESP32-CAM struggled in low-light conditions, affecting image quality.

- Component Integration: Synchronizing the operation of the LCD, buzzer, sensor, and camera with the ESP32 was tricky.

- WiFi Connectivity:Frequent disconnects affected real time data transmission and image upload.

FUTURE SCOPE

- Integrate with municipal systems to monitor noise pollution in urban areas and assist in enforcing regulations.
- If unusual or loud noises (like shouting, glass breaking) are detected, trigger an emergency alert for security purposes.
- Connect with smart traffic systems to monitor and control honking or loud vehicles in restricted zones.

CONCLUSION

The development of the Noise Pollution Detection System successfully demonstrates how IoT technology can be applied to monitor and control noise levels in sensitive environments. By integrating ESP32, ESP32-CAM, sound sensors, and wireless communication, the system provides real-time alerts and visual monitoring, ensuring prompt response to excessive noise. This project not only helps in maintaining a quiet atmosphere in places like industries,offices, classrooms and libraries but also promotes awareness about noise pollution. With further enhancements, such systems can be expanded for use in public areas and smart city applications.

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THANK YOU